

7. SYSTEM DESCRIPTION

The Long Range Fuel System provides for additional storage of up to 400 liters fuel when both aux (auxiliary) fuel tanks are installed in the cargo compartment. The system fixed provisions comprise a LONG (LG) RANGE TANK control panel (mounted either on the instrument panel or center console), main fuel tank connections, electrical system connections and tank mounting rails.

Each 200-liter capacity aux tank is fitted with a filler neck, a float switch, an electrical valve, vent and drain lines, a fuel transfer line, and electrical wiring. A sidewall protective panel shall be fitted on single-tank installations to prevent damage from cargo.

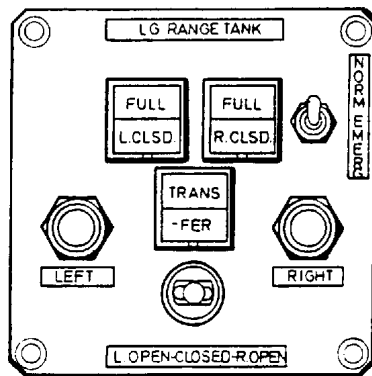
Each aux tank has two indicator lights on the control panel; FULL indicating fuel quantity is approx 185 liters, and L. CLSD (R. CLSD) indicating that the respective tank valve is closed. Valve position indication is also provided by a valve-integrated mechanical indicator that, depending upon valve manufacturer, will point to "O" or show red when the valve is open.

Fuel from the aux tanks is gravity transferred to the helicopter main fuel tank by opening each valve. These valves are operated from a single (L.) OPEN-CLOSED-(R.) OPEN selector switch on the control panel so that only fuel from one tank may be transferred at a time. The transfer operation is automatic provided a tank valve is selected for opening before the **TRANSFER** caution light comes on.

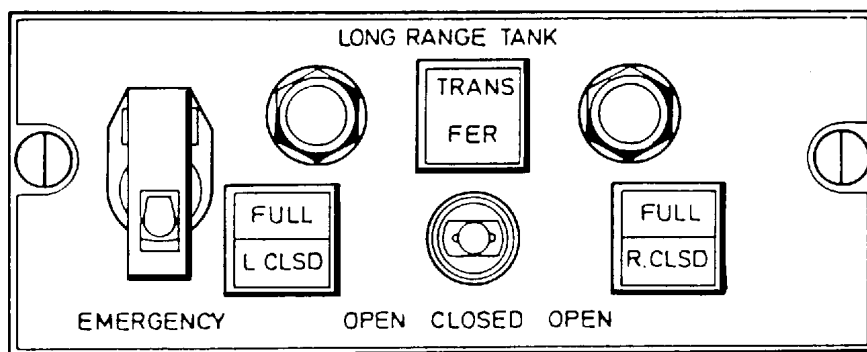
Although a tank valve is selected OPEN, the valve will not open, however, until the main tank float switch signals that the main tank can accept fuel transfer (**TRANSFER** caution light comes on).

An EMERGENCY (NORM-EMERG) switch is provided to override the main tank float switch and open the selected tank valve before the **TRANSFER** caution light comes on.

The LONG (LG) RANGE TANK control panel circuit breakers (LEFT and RIGHT) are supplied 28V DC direct from the main bus.



CONTROL PANEL (INSTRUMENT PANEL INSTALLATION)



CONTROL PANEL (CENTER CONSOLE INSTALLATION)

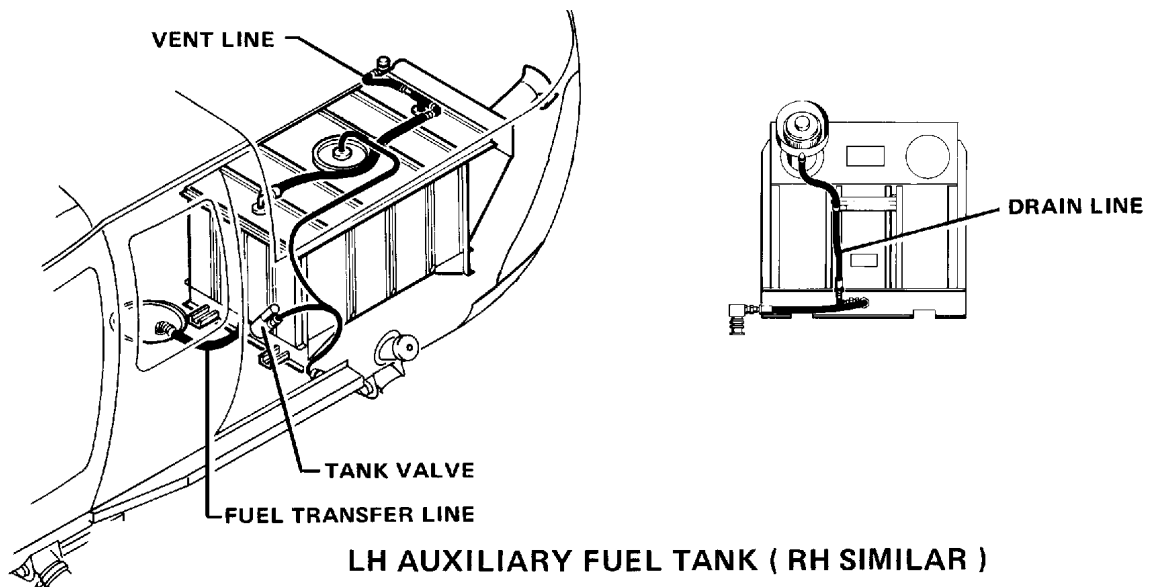


Fig. 1 Long Range Fuel System